Badreddine Benfettouma

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Research interests

Digital integrated-circuit design and RISC-V microarchitecture; formal verification of hardware; energy-efficient hardware for machine-learning inference; open-source EDA flows and open PDKs.

Education

Licence 3 (third year of the Bachelor), Electronics 2026–2027
Université Bourgogne Europe, Dijon, France

Licence 1 and 2, Computer Science and Electronics track 2024–2026
Université de Bourgogne (Université Bourgogne Europe since January 2025), Dijon, France
Selected grades (out of 20): computer architecture 14.2 (lab 18.1), signal processing 17.5, scientific computing 20.

The French *Licence* is a three-year bachelor's degree (180 ECTS).

Projects

RV32I-Alpha: a two-stage RISC-V processor core 2026
VHDL, SystemVerilog; source code

- Two-stage pipelined core implementing the 37 instructions of the RV32I base ISA, with traps on illegal instructions and misaligned accesses and a data-memory port that accepts wait states. Written in VHDL, then ported to SystemVerilog.
- Formal verification: riscv-formal instruction and consistency checks (bounded model checking, 10–15 cycles) and custom SymbiYosys properties for the memory ports; mutation testing (8 injected bugs, all detected); Yosys proof that the VHDL and SystemVerilog versions are sequentially equivalent.
- FPGA: Digilent Basys 3 board (Artix-7); the core alone uses 675 LUTs and meets 100 MHz (Vivado).
- ASIC: hardened to a DRC- and LVS-clean GDSII macro in SkyWater SKY130 with LibreLane (power grid, clock tree, sign-off; 0.22 mm^2, 40 MHz at the typical corner).

Area and energy accounting of logic-gate neural networks 2026
Independent project, write-up in progress; Python, PyTorch, Yosys, OpenROAD, OpenSTA

- Built an open flow that takes a trained logic-gate network to a placed SKY130 netlist, simulates it at gate level on real test images and derives power from the recorded switching activity.
- Checked that the gate-level netlist reproduces the software model bit for bit. Included the cost of input buffering, output logic and wiring, which published comparisons usually leave out.

Technical skills

HDLs	VHDL, SystemVerilog, Verilog
Verification	SymbiYosys, riscv-formal, Yosys equivalence checking, GHDL, Icarus Verilog
Implementation	Vivado (Artix-7); LibreLane, Yosys, OpenROAD, OpenSTA, Magic, KLayout
Circuits	ngspice with SKY130 device models, Monte-Carlo mismatch analysis
Programming	Python (NumPy, PyTorch), Tcl, RISC-V assembly; Linux, Git

Languages

French	native
English	C1
Japanese	learning; level certification in progress
Arabic	Algerian Arabic (spoken); reads the Arabic script